Université de Sherbrooke
Summer school in biodiversity modelling   August 26th, 2026

Metacommunities, distribution and competition

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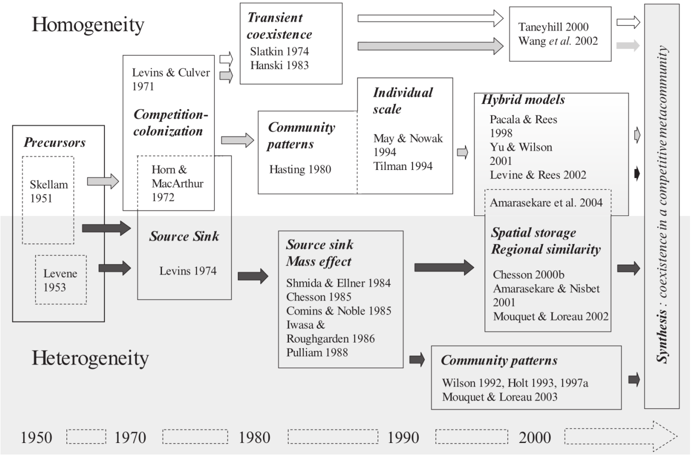
Recall



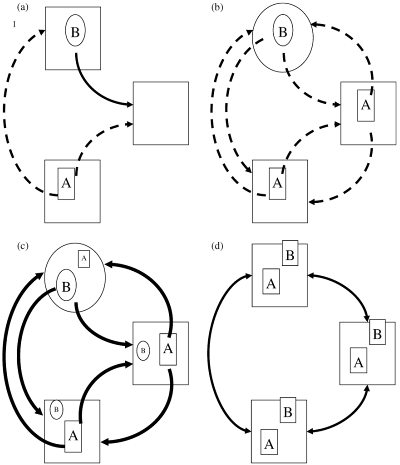
Mouquet et al. 2005 in Holyoak et al. Metacommunity Ecology. Chicago University Press

Definition

A set of local communities that are linked by dispersal of multiple potentially interacting species

Leibold et al. 2004. Eco. Lett

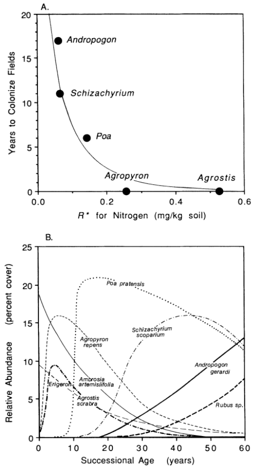
Definition



Leibold et al. 2004. Eco. Lett

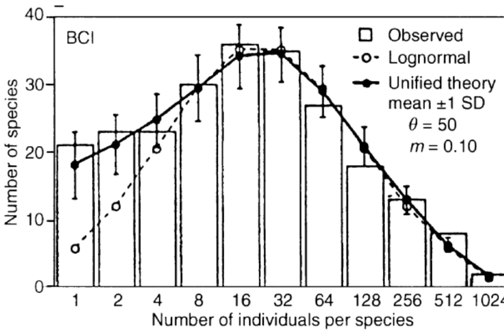
Patch dynamics

A perspective that assumes that patches are identical and that each patch is capable of containing populations. Patches may be occupied or unoccupied. Local species diversity is limited by dispersal. Spatial dynamics are dominated by local extinction and colonization



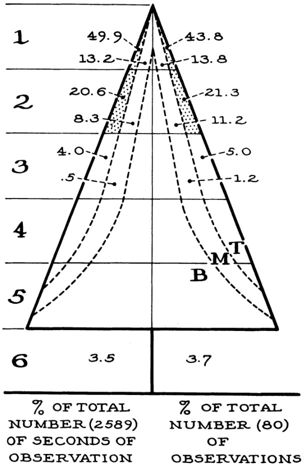
Neutral theory

A perspective in which all species are similar in their competitive ability, movement and fitness (Hubbell 2001). Population interactions among species consist of random walks that alter relative frequencies of species. The dynamics of species diversity are then derived both from probabilities of species loss (extinction, emigration) and gain (immigration, speciation).



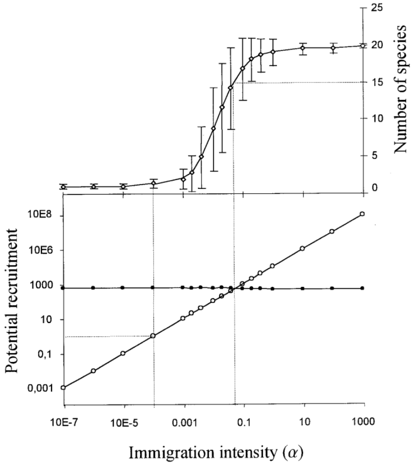
Species sorting

A perspective that emphasizes the resource gradients or patch types cause sufficiently strong differences in the local demography of species and the outcomes of local speciesÕ interactions that patch quality and dispersal jointly affect local community composition. This perspective emphasizes spatial niche separation above and beyond spatial dynamics. Dispersal is important because it allows compositional changes to track changes in local environmental conditions



Mass effect

A perspective that focuses on the effect of immigration and emigration on local population dynamics. In such a system species can be rescued from local competitive exclusion in communities where they are bad competitors, by immigrate from communities where they are good competitors. This perspective emphasizes the role that spatial dynamics affect local population densities.



Discussion

Neutral model Verbal description

- Consider a lattice with each cell occupied by a single individual
- Time scale corresponds to the replacement of 1 individual at a time
- All species have the same probability of mortality
- Compute number of seeds reaching a seed trap at the center of the cell
- Dispersal occurs from neighbours with probability (1-m) and from a regional pool with probability m
- For simplicity, all species have the same abundance in the regional pool
- All species produce the same number of seeds
- Recruitment occurs by selecting a seed at random from the seed trap
- Repeat for a (very) large number of time steps

Neutral model Pseudo-code

```
SET run parameters $l$, $s$, $nsteps$
SET ecological parameters $m$ and $r$

DEFINE array XY of spatial coordinates of size $l^2$
DEFINE array J of size $l^2 x S$

COMPUTE connectivity matrix ConMat

FOR n in 1:nsteps
    SELECT a cell z at random
        J[z] = 0
    COMPUTE replacement probability for each species
   DRAW a species identity using multinomial distribution
    UPDATE array J
```

```
# Local replacement
    n_propagules = ContMat%*%J
    p_local = n_propagules/(r^2-1)

# Regional replacement
    p_regional = 1/s

# Final probability
    p = (1-m)*p_local + m*p_regional
```

Add species differences

- Consider that each cell has a unique environmental condition between 0 and 1
- The local environment impacts seed germination
- Response to the environment is unimodal
- Each speices has a niche optimum between 0 and 1
- All species have the same niche width

```
# Local replacement
    n_propagules = ContMat%*%J
    survival = survival_fn(E,O,sigma)
    p_local = survival * n_propagules/(r^2-1)
```

Compare expectations

Simulations

- Local abundance distribution at equilibrium
- Regional abundance distribution at equilibrium
- Species area relationship
- Shared species as a function of distance
- Species-environment relationship

Advanced

- Compare different spatial autocorrelations to evaluate the effect of mass effect
- Think about a way to incorporate competition-colonization trade-off

Advanced Other topics

- Null hypothesis or a baseline ?
- Discriminating paradigms
- Quasi-neutral models
- Continuous vs discrete organization
- Spatial network of patches
- Spatial contingency
- Disturbances
- Speciation model (point mutation, random fission, protracted, reproductive isolation)
- Phylogeny
- From genes to communities
- Moments of metacommunities